Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 06



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Recap of Previous Lecture



- Topic — Candidate key
- Topic — Primary key
- Topic — Alternate key
- Topic — Super key

Slide

Topics to be Covered



- **Topic** Super key
- **Topic** Closure of an attribute set
- **Topic** Identification of candidate keys in a relation

Slide

Key Concept :-

In a relational table no two tuples of the table can be exactly same

↳ i.e, in relational table duplicate tuples are not allowed.

↳ To implement this restriction we say that every relation must have a key.

* A key in a relation is the set of attributes that can uniquely identify each tuple of the relation



Topic : Different types of keys

There are various types of keys

1. Candidate key
2. Primary key
3. Alternate key
4. Super key
5. Foreign key

Foreign key Concept will be discussed in the last Capter i.e: "ER Model & Integrity Constraints"

* **Minimal Set** :→ A set of elements from which no element can be removed without loosing the associated property is called a minimal set.

i.e. whenever we remove any element from the set then it is guaranteed that it will lose its property

Note :→ If values of a set of attributes are guaranteed to be unique in a relation, then that set of attributes is definitely a "key" of that relation {But it may or may not be minimal}



Topic : Candidate key

i.e. no attribute can be removed from the set without destroying its property of being a key {i.e. without destroying its property of being unique}

A minimal set of attributes that can identify each tuple of the relation uniquely is called a Candidate key

(or)

A minimal set of attributes that can determine each attribute of the relation is called a Candidate key.

eg①:- Consider the following relation

Student

Sid	Sname	Fee
S_1	A	500
S_2	A	500
S_3	B	600
S_4	B	400
S_5	C	600

Also consider that following functional dependencies holds in the relation

$\{ Sid \rightarrow Sname \}$
$\{ Sid \rightarrow fee \}$

$Sid \rightarrow Sname, fee, Sid$

* With respect to given FDs we can observe that values of Sid will always be unique in the relation Student Hence, "Sid" will be a key of the relation "Student"
* If values of Sid are unique, then values of (Sid, Sname) together will also be unique
Hence, (Sid, Sname) is also a key of "Student"

eg②: Consider the following relation

Enroll

Sid	Cid	I-id
S_1	C_1	101
S_2	C_1	101
S_3	C_2	101
S_3	C_3	102
S_4	C_3	102

Consider following functional dependency holds in the relation

$\{ Cid \rightarrow I\text{-}id \}$
$\Downarrow$
$Sid, Cid \rightarrow Sid, I\text{-}id$
$\Downarrow$
$Sid, Cid \Rightarrow Sid, Cid, I\text{-}id$

* Together the values of (Sid, Cid) will be unique in the relation, Hence (Sid, Cid) is a key of the relation Enroll.
* Values of (Sid, Cid) are unique, ∴ Values of (Sid, Cid, I-id) will also be unique.
Hence, (Sid, Cid, I-id) is also a key of relation "Enroll"

eg①:- Consider the following relation

Student

Sid	Sname	Fee
S_1	A	500
S_2	A	500
S_3	B	600
S_4	B	400
S_5	C	600

Also consider that following functional dependencies holds in the relation

{ Sid → Sname
Sid → fee }

Sid → Sname, fee, Sid

* With respect to given FDs we can observe that values of Sid will always be unique in the relation Student Hence, "Sid" will be a key of the relation "Student"
* If values of Sid are unique, then values of (Sid, Sname) together will also be unique
Hence, (Sid, Sname) is also a key of "Student"

In eg①:-

(Sid, Sname) is a key of the relation Student, but it is not minimal, Because if we remove "Sname" from the set then "Sid" alone can uniquely identify each tuple of the relation

∴ (Sid, Sname) is not a Candidate key of the relation

In eg① :- (Sid) is also a key of the relation, and it is minimal as well,

∴ (Sid) is a Candidate key of relation Student

eg②: Consider the following relation

Enroll

Sid	Cid	I-id
S_1	C_1	101
S_2	C_1	101
S_3	C_2	101
S_3	C_3	102
S_4	C_3	102

Consider following functional dependencies holds in the relation

$\{ Cid \longrightarrow I\text{-}id \}$

$\Downarrow$

$Sid, Cid \longrightarrow Sid, I\text{-}id$

$\Downarrow$

$Sid, Cid \Rightarrow Sid, Cid, I\text{-}id$

- Together the values of (Sid, Cid) will be unique in the relation,
  Hence (Sid, Cid) is a key of the relation Enroll.
- Values of (Sid, Cid) are unique, so Values of (Sid, Cid, I-id) will also be unique.
  Hence, (Sid, Cid, I-id) is also a key of relation "Enroll"

In eg②:

(Sid, Cid) is a key of the relation Enroll.

- If we remove Sid from the set, then "Cid" alone can not uniquely identify each tuple of relation.

and if remove "Cid" from the set, then "Sid" alone can not uniquely identify each tuple of the relation.

i.e. no attribute can be removed from the set without destroying its property of being a key

so (Sid, Cid) is a minimal key and Hence a Candidate key of relation "Enroll".

- In eg ②:- (Sid, Cid, I-id) is also a key of relation, but it is not minimal because "I-id" can be removed from the set without destroying its property of being a key.

∴ (Sid, Cid, I-id) is not a Candidate key.



Topic : Candidate key

* If a key is formed of a single attribute then it is always minimal, and hence a candidate key

① If Candidate key is formed of a single attribute. then such candidate keys are called simple Candidate key

② If Candidate key is formed of two or more attributes then such Candidate keys are called Compound or Composite Candidate key.

③ A relation may have more than one candidate key.
{The number of candidate keys in a relation will be based on the functional dependencies that holds in that relation}



Topic : Candidate key

④ Attributes belonging to any of the Candidate key of the relation are called "prime attributes" of the relation.

⑤ Attributes that does not belong to any of the Candidate key are called Non-prime attributes.



Topic : Primary key

- There may be multiple candidate keys in a relation. And one of those candidate key may be defined as "primary key" of that relation.
- In a relation we can have at most one primary key.
- Attributes of primary key are not allowed to take "NULL" values.



Topic : Alternate key

{ Also known a Secondary key }

All candidate keys Except primary key are Called atlernate keys



Topic : Super key

"need not be minimal"

A set of attributes that uniquely identify each tuple of the relation is called a super key.

(or)

A set of attributes (need not be minima) that can determine all attributes of the relation is called a Super key.

* Any super-set of a candidate key is a Super key.
* Every candidate key is also a Super key, but Every Super key need not be a candidate key.

Note:- At least one subset of every Super key is guaranteed to be a Candidate key.

Note:-

$$\text{Super key} = (\text{Attributes of any one of the Candidate key}) + (\text{0 or more attributes out of remaining attributes})$$

eg: Consider the following relational schema

Student (Sid, Sname, Branch)

If "Sid" is the only candidate key of the relation Student, then identify the total number of superkeys in relation Student.

* Any superset of {Sid} will be a Superkey

Supersets of {Sid} =
- {Sid} } C.K + 0 out of remaining two
- {Sid, Sname} } C.K + 1 attribute out of remaining two
- {Sid, Branch} } C.K + 1 attribute out of remaining two
- {Sid, Sname, Branch} } C.K + 2 out of remaining two

← This are the 4 superkeys of relation Student.

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$

Find the number of superkeys possible in relation R.

(i) When "A_1" is the only candidate key of relation R.

Super Key of Relation R = attribute A_1 must be present (and) { 0 or more attributes out of remaining (n-1) attributes }

Number of Superkeys = $1 * \{ {}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \ldots\ldots\ldots + {}^{(n-1)}C_{(n-1)} \}$

A_1 can be selected only in '1' way

$= 1 * 2^{n-1}$

$= 2^{n-1}$ Ans

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots, A_n)$

Find the number of superkeys possible in relation R.

(i) When "A_1" is the only candidate key of relation R.

Super Key of Relation R = attribute A_1 must be present and (other attributes may or may not be present)

Number of Super keys = A_1 A_2 A_3 A_4 A_5 - - - - A_n

$1 * 2 * 2 * 2 * 2 * \ldots * 2 = 2^{n-1}$

$$ {}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots + {}^{n}C_n = 2^n $$

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$

Find the number of superkeys possible in relation R.

(ii) When (A_1A_2) is the only candidate key of relation R.

Super Key = (A_1 & A_2 must be present) and other attributes may or may not be present

A_1 A_2 A_3 A_4 ..... A_n

Number of Super keys = $1 * 1 * 2 * 2 * \ldots\ldots * 2 = 2^{n-2}$

$= 2^{n-2}$

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots, A_n)$

Find the number of superkeys possible in relation R.

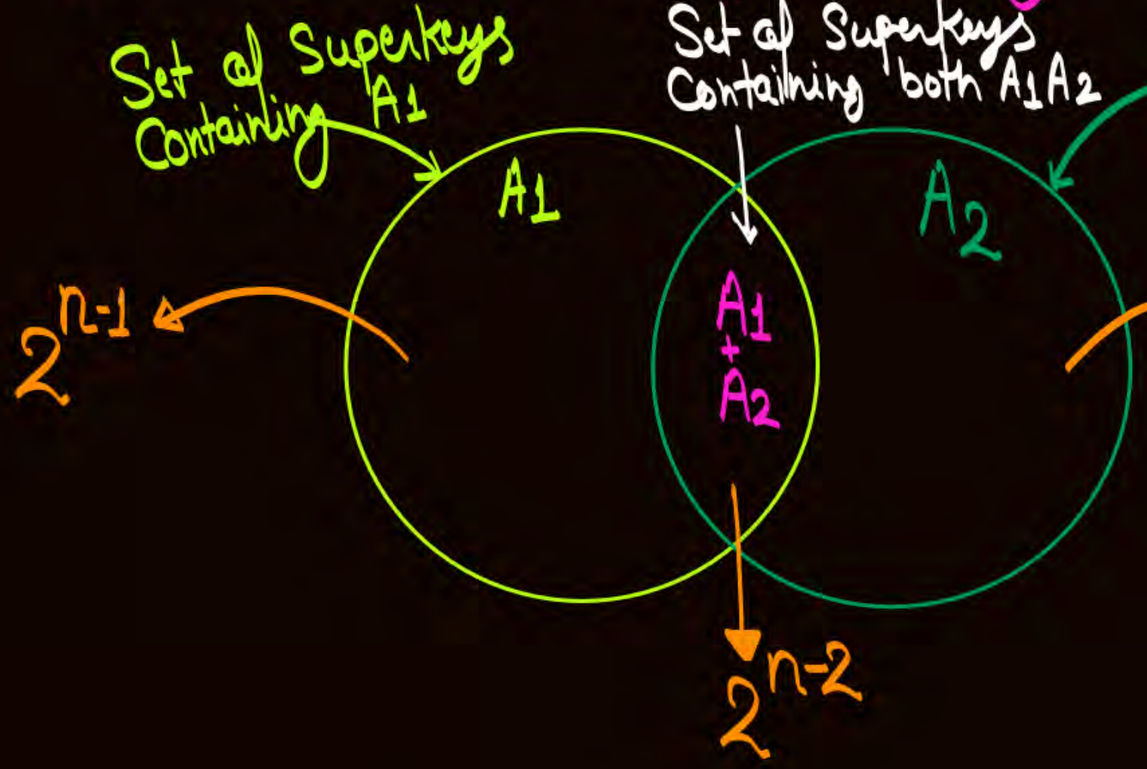
(iii) When "A_1" & "A_2" are the only two candidate keys of relation R.

	A_1	A_2	A_3	A_4 ..... A_n	
Super key =	1✓	1×	2	2 ... 2	$= 2^{n-2}$
(or)					+
	1×	1✓	2	2 ... 2	$= 2^{n-2}$
(or)					+
	1✓	1✓	2	2 2	$= 2^{n-2}$
					$3 \cdot 2^{n-2}$

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$

Find the number of superkeys possible in relation R.

(iii) When "A_1" & "A_2" are the only two candidate keys of relation R.



$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$n(A_1 \text{ or } A_2) = n(A_1) + n(A_2) - n(A_1 \text{ and } A_2)$

$= 2^{n-1} + 2^{n-1} - 2^{n-2}$

$= 2^1 * 2^{n-2} + 2^1 * 2^{n-2} - 2^{n-2}$

$= 4 \cdot 2^{n-2} - 2^{n-2}$

$= 3 \cdot 2^{n-2}$

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots, A_n)$

Find the number of superkeys possible in relation R.

(iv) When (A_1A_2) and (A_3A_4) are two candidate keys of relation R.

$(A_1\ A_2)\ (A_3\ A_4)\ A_5\ A_6 \ldots A_n$

$-1 * -1 * \times 1 * \times 1 * 2 * 2 * \ldots 2 = 2^{n-4}$

$-1 * -1 * -1 * \times 1 * 2 * 2 * \ldots 2 = +2^{n-4}$

$-1 * -1 * \times 1 * -1 * 2 * 2 * \ldots 2 = +2^{n-4}$

$-1 * -1 * -1 * -1 * 2 * 2 * \ldots 2 = +2^{n-4}$

$\times 1 * \times 1 * -1 * -1 * 2 * 2 * \ldots 2 = +2^{n-4}$

$-1 * \times 1 * -1 * -1 * 2 * 2 * \ldots 2 = +2^{n-4}$

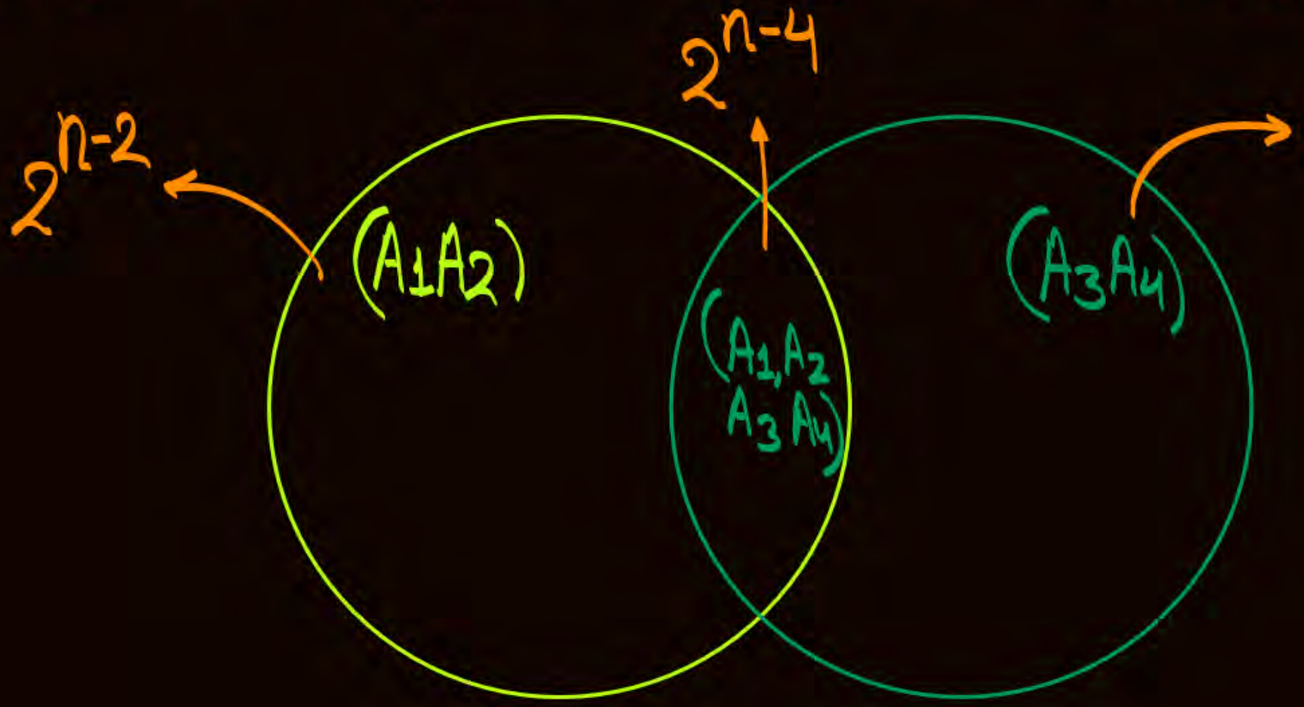
$\times 1 * -1 * -1 * -1 * 2 * 2 * \ldots 2 = +2^{n-4}$

$7 * 2^{n-4}$ Ans

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$

Find the number of superkeys possible in relation R.

(iv) When (A_1A_2) and (A_3A_4) are two candidate keys of relation R.



Total no. of Super keys $= 2^{n-2} + 2^{n-2} - 2^{n-4}$

$= 2^2 \cdot 2^{n-4} + 2^2 \cdot 2^{n-4} - 2^{n-4}$

$= 4 \cdot 2^{n-4} + 4 \cdot 2^{n-4} - 2^{n-4}$

$= 8 \cdot 2^{n-4} - 2^{n-4}$

$= 7 \cdot 2^{n-4}$ Ans

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$

Find the number of superkeys possible in relation R.

(V) When (A_1A_2) & (A_2A_3) are only two candidate keys of relation R.

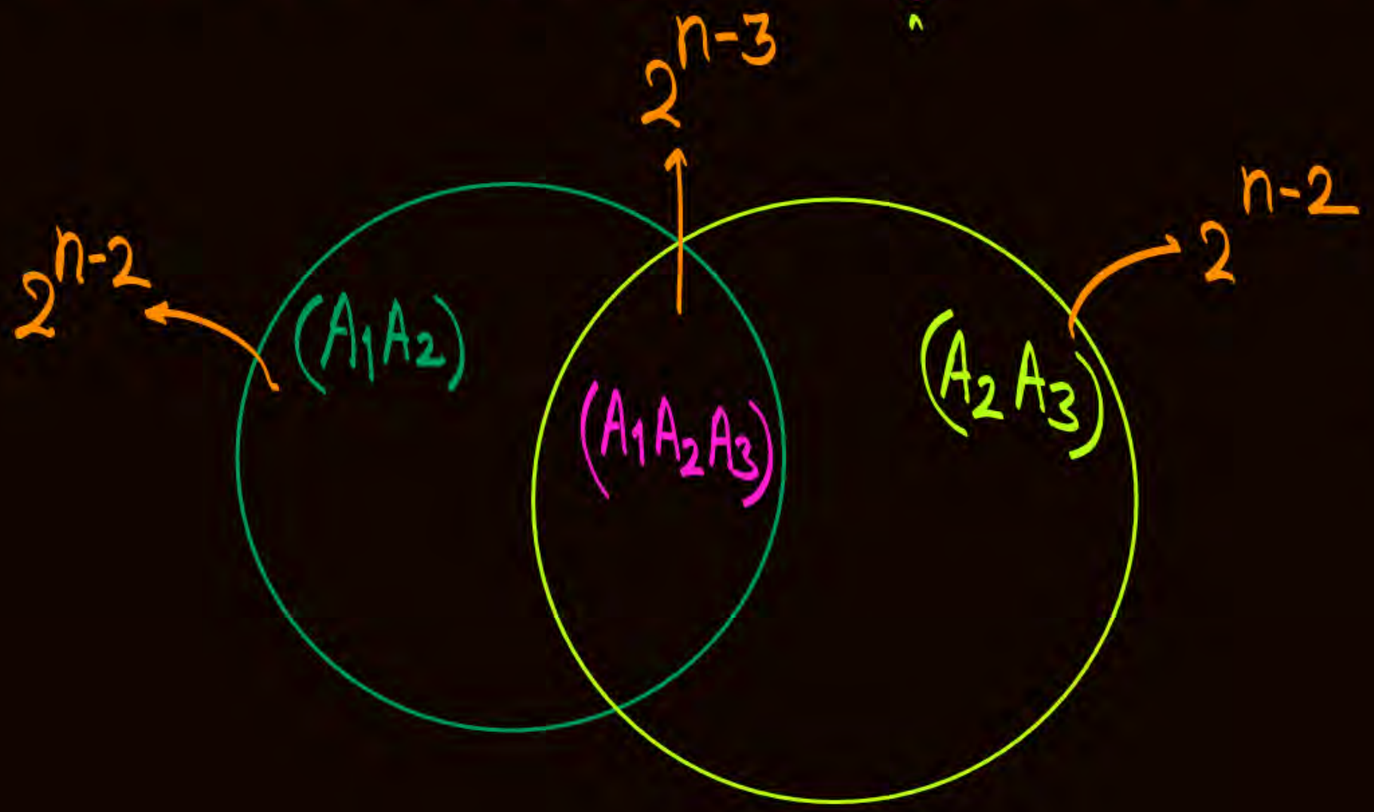
A_1	A_2	A_3	A_4	A_5 - - -	A_n	
✓1 *	✓1 *	×1 *	2 *	2 * - - -	2	$= 2^{n-3}$
×1 *	✓1 *	✓1 *	2 *	2 * - - -	2	$+ 2^{n-3}$
✓1 *	✓1 *	✓1 *	2 *	2 * - - -	2	$+ 2^{n-3}$

$$= 3 \cdot 2^{n-3}$$

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$

Find the number of superkeys possible in relation R.

(V) When (A_1A_2) & (A_2A_3) are only two candidate keys of relation R.



Total no. of Super keys $= 2^{n-2} + 2^{n-2} - 2^{n-3}$

$= 3 \cdot 2^{n-3}$

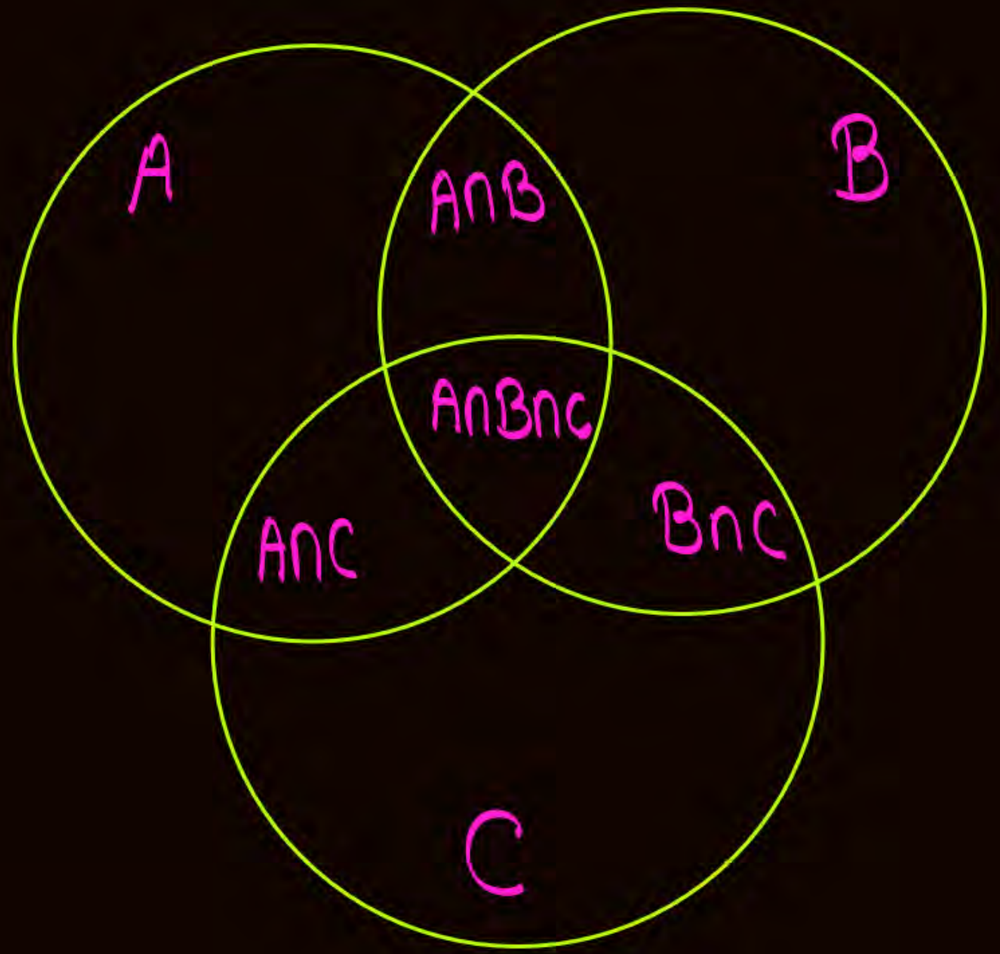
Ans

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$

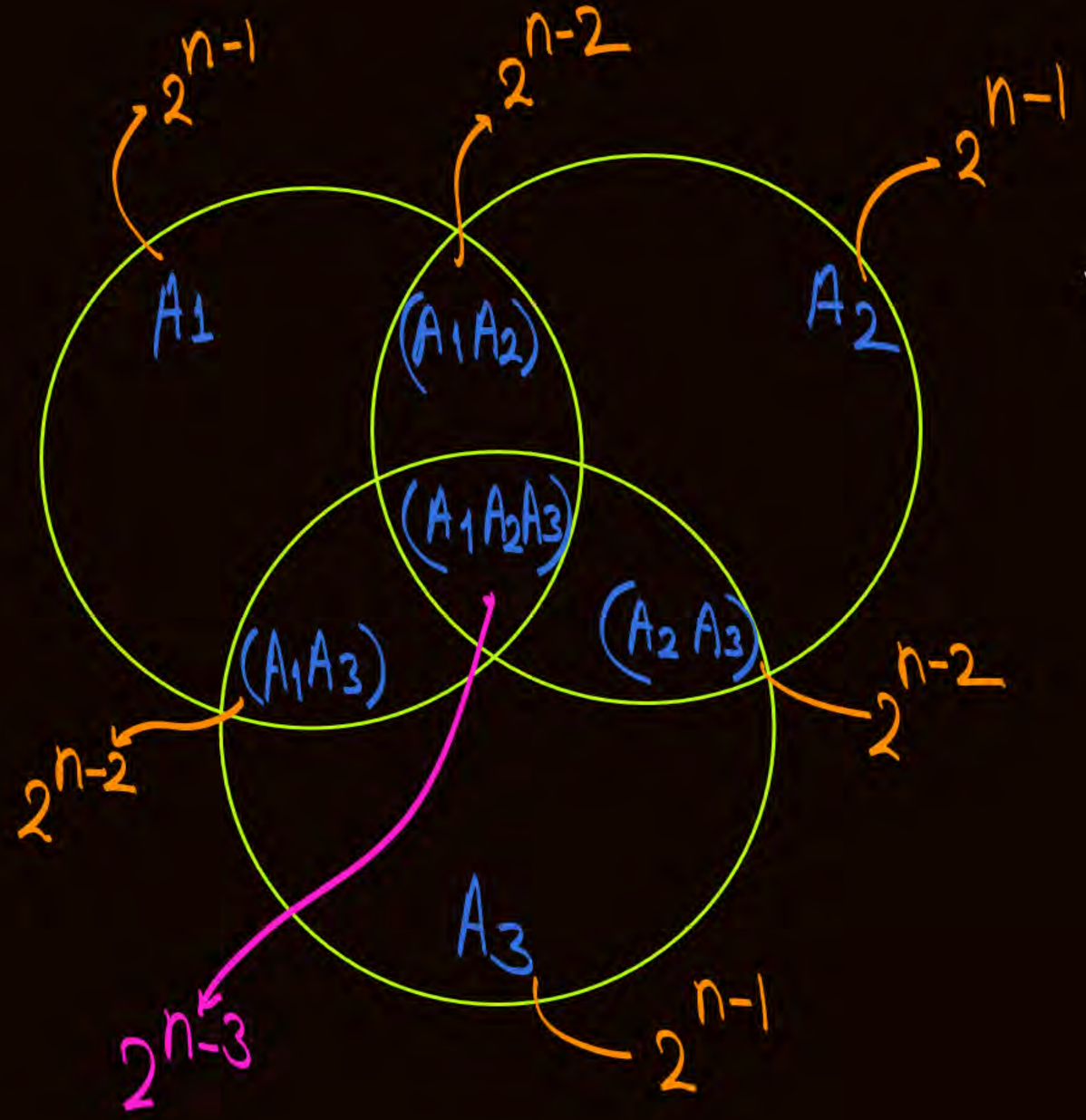
Find the number of superkeys possible in relation R.

(vi) When (A_1), (A_2) and (A_3) are three candidate keys of relation R.

Soluⁿ is on next to next slide



$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$



Total no. of Superkeys $= 2^{n-1} + 2^{n-1} + 2^{n-1} - 2^{n-2} - 2^{n-2} - 2^{n-2} + 2^{n-3}$

$= 7 \cdot 2^{n-3}$ Ans

Q:- Consider the relational schema $R(A_1, A_2, A_3, \ldots\ldots, A_n)$ Find the number of superkeys possible in relation R. When each of relation R itself is a Candidate Key

Ans $= (2^n - 1)$

2^n: Any subset

1: Except Empty subset



Topic : Closure of an attribute set

- Closure of an attribute set X (i.e., X^+) can be defined as set of all the attributes that can be functionally determined from attributes of set X.

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

A→B

B→C

C→D

$(A)^+ = \{A, B, C, D\}$

$(AB)^+ = \{A, B, C, D\}$

$(AC)^+ = \{A, B, C, D\}$

$(AD)^+ = \{A, B, C, D\}$

$(ABC)^+ = \{A, B, C, D\}$

$(ABD)^+ = \{A, B, C, D\}$

$(ACD)^+ = \{A, B, C, D\}$

$(ABCD)^+ = \{A, B, C, D\}$

$(B)^+ = \{B, C, D\}$

$(BC)^+ = \{B, C, D\}$

$(BD)^+ = \{B, C, D\}$

$(BCD)^+ = \{B, C, D\}$

$(C)^+ = \{C, D\}$

$(CD)^+ = \{C, D\}$

$(D)^+ = \{D\}$

#e.g. Consider the following FD set

$F = \{ AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A\}$

find the closure of following set of attributes.

(i) $\{C,F\}^+ = \{C, F, G, E, A, D\}$

(ii) $\{B,G\}^+ = \{B, G, A, C, D\}$

(iii) $\{A,F\}^+ = \{A, F, D, E\}$

(iv) $\{A,B\}^+ = \{A, B, C, D, G\}$



Topic : Super key

- Let R be the relational schema, and let X be some set of attributes over relation R. If X^+ Contains all attributes of relation R, then X is called super key of relation R.

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

$(A)^+ = \{A, B, C, D\}$

$(AB)^+ = \{A, B, C, D\}$

$(AC)^+ = \{A, B, C, D\}$

$(AD)^+ = \{A, B, C, D\}$

$(ABC)^+ = \{A, B, C, D\}$

$(ABD)^+ = \{A, B, C, D\}$

$(ACD)^+ = \{A, B, C, D\}$

$(ABCD)^+ = \{A, B, C, D\}$

Closure Contains all attributes of the relation.

∴ All eight sets of attributes are Super keys of relation R

$(B)^+ = \{B, C, D\}$

$(BC)^+ = \{B, C, D\}$

$(BD)^+ = \{B, C, D\}$

$(BCD)^+ = \{B, C, D\}$

$(C)^+ = \{C, D\}$

$(CD)^+ = \{C, D\}$

$(D)^+ = \{D\}$

Proper subset :- For a given set A, any subset of set A, except set A itself is called Proper subset of set A.

eg: let $A = \{a, b, c\}$

Subsets of Set A are = $\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}$

$\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}$ — Proper Subsets of $\{a,b,c\}$

$\{a,b,c\}$ — it is equal i.e. Not a Proper Subset



Topic : Candidate key (Minimal Super key)

- Let R be the relational schema, and let X be the super key of relation R.
- If no proper subset of X is a super key the X is minimal super key i.e., X is Candidate key

eg: let $(A,B)^+$ = { All attributes of Relⁿ R }
∴ AB is a Super key of R

Non-Empty Proper subsets of {A,B} are

{A}, let $(A)^+$ = Not all attributes of R
∴ A is not a S.K.

{B}, Let $(B)^+$ = Not all attributes of R
∴ B is not a S.K

No Proper subset of {A,B} is a Superkey
∴ (AB) is minimal S.K
Hence, (AB) is C.K

H.W

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

Find the Candidate key of R.

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

E→A

Find the Candidate key of R.

#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

AB→CD,

D→A

Find all the Candidate keys of R.

#Q. Assume a relation R (M, N, O, P, Q) that has the following functional dependencies:

MNO $\rightarrow$PQ and

P $\rightarrow$ MN

Find the Candidate keys of R.

#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

$AB \rightarrow CD$

$C \rightarrow A$

$D \rightarrow B$

Find the Candidate keys of R.

#Q. Assume a relation R (A, B, C, D, E, H) that has the following functional dependencies:

$A \rightarrow B$

$BC \rightarrow D$

$E \rightarrow C$

$D \rightarrow A$

Find the Candidate keys of R.



2 mins Summary

- Topic: Super key
- Topic: Closure of an attribute set
- Topic: Identification of candidate keys in a relation

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THANK - YOU

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